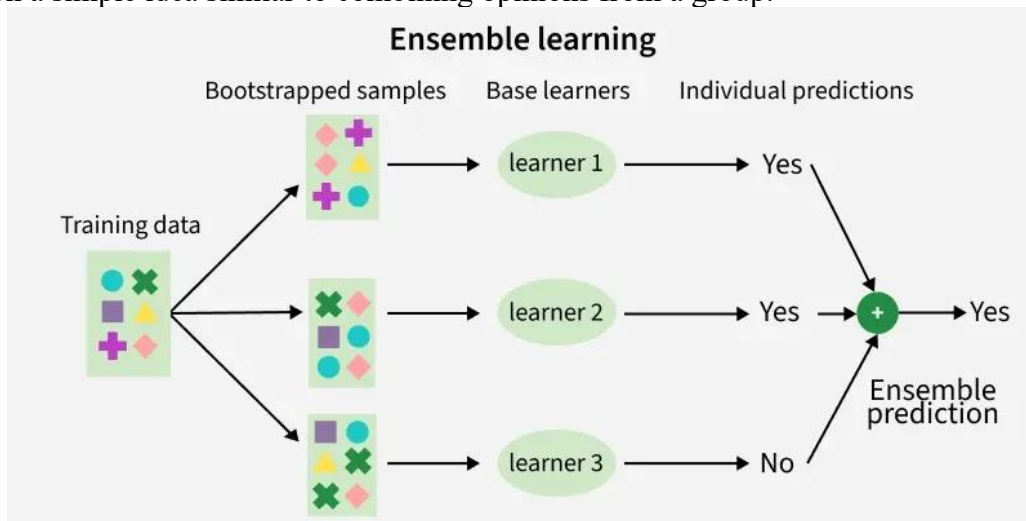


Q1. What is Ensemble Learning technique? What are the types of it?

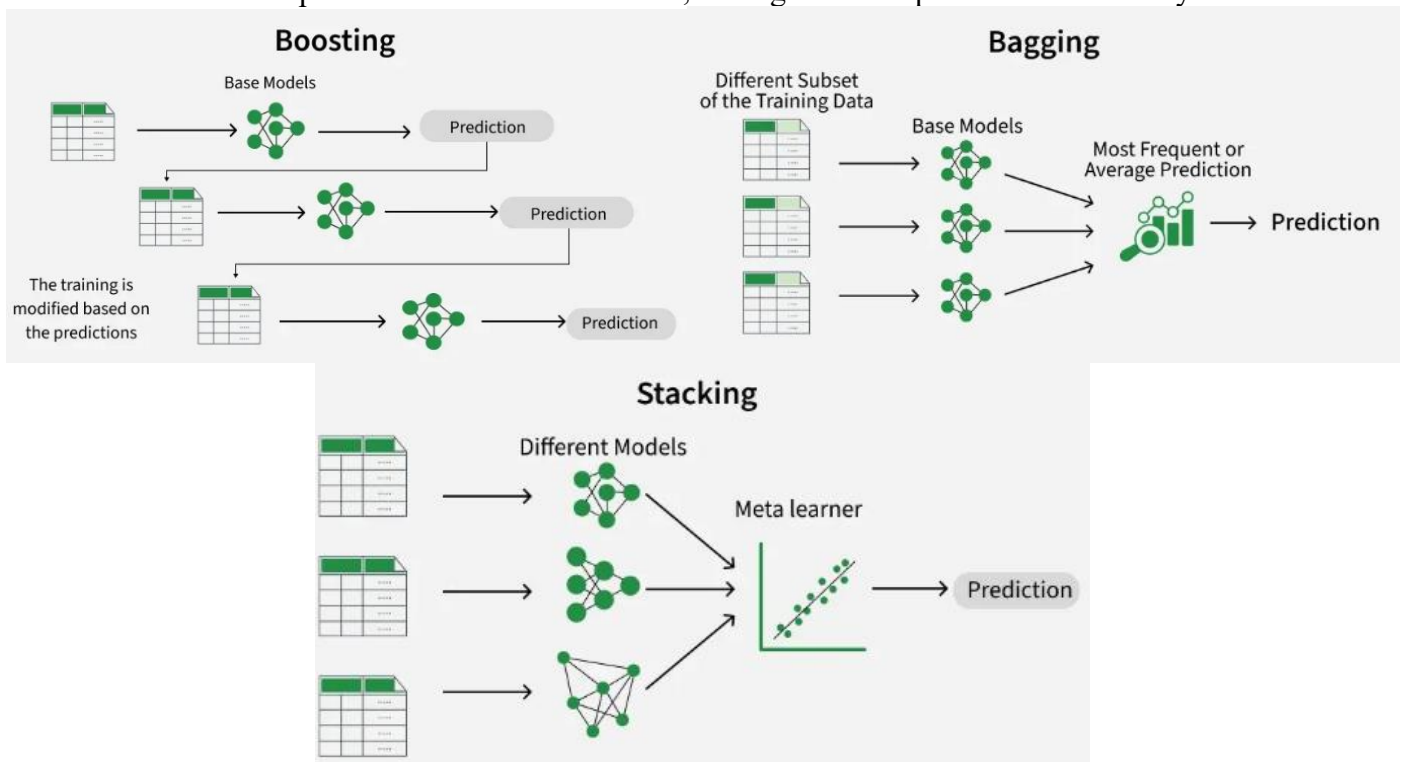
Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.

- Uses multiple models together to improve overall accuracy.
- Reduces errors by balancing mistakes across models.
- Works on a simple idea similar to combining opinions from a group.



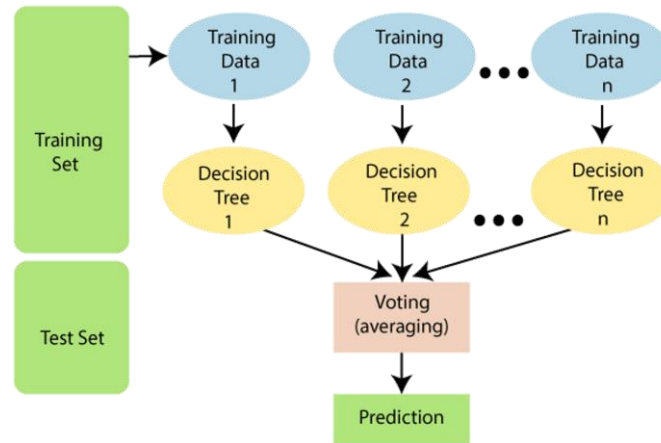
Types of Ensemble Learning

1. **Bagging (Bootstrap Aggregating):** Models are trained independently on different random subsets of the training data. Their results are then combined—usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.
2. **Boosting:** Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones. The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.
3. **Stacking (Stacked Generalization):** Multiple different models (often of different types) are trained and their predictions are used as inputs to a final model, called a meta-model. The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.



Q2. What is Random Forest Algorithm? How does it work? Explain with a real-life example.

Random Forest is a classifier that comprises a number of decision trees on various subsets of the provided dataset and takes the average to enhance the predicted accuracy of that dataset. Instead of depending on a single decision tree, the random forest collects the predictions from each tree and predicts the final output based on the majority vote of predictions.



Why use Random Forest?

Here are some reasons why we should utilize the Random Forest algorithm:

- It requires shorter training time than other algorithms.
- It predicts output with great accuracy, and it works efficiently even on big datasets.
- It can also retain accuracy when a significant amount of data is absent.

How does the Random Forest Algorithm Work?

Random Forest operates in two stages: the first is to generate the random forest by mixing N decision trees, and the second is to make predictions for each tree generated in the first phase.

Step 1: Choose K data points at random from the training set.

Step 2: Create decision trees for the specified data points (Subsets).

Step 3: Determine the number N for the number of decision trees you wish to construct.

Step 4: Repeat Steps 1 and 2.

Step 5: Find the predictions of each decision tree for new data points and assign the new data points to the category with the most votes.

The following random forest algorithm example will help you understand how the algorithm works:

Assume there is a dataset containing several fruit photos. As a result, the Random Forest classifier is given this dataset. The dataset is subdivided and distributed to each decision tree. During the training phase, each decision tree gives a prediction result, and when a new data point occurs, the Random Forest classifier predicts the final choice based on the majority of outcomes. Consider the following image:

